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Studierichting: Informatica
Oefeningenreeks 4A nr. 49.8

$$\begin{aligned} & \int \frac{\sin(2x)}{1 + \sin^2(x)} dx \\ &= \int \frac{\sin(2x)}{1 + \frac{1 - \cos(2x)}{2}} dx \\ &= 2 \int \frac{\sin(2x)}{3 - \cos(2x)} dx \\ &u = 3 - \cos(2x) \Rightarrow du = 2 \sin(2x) dx \\ &= \int \frac{du}{u} \\ &= \ln |3 - \cos(2x)| + c \end{aligned}$$

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$$\begin{aligned} & \int \frac{\sin(2x)}{1 + \sin^2(x)} dx \\ &= \int \frac{2 \sin(x) \cos(x)}{1 + \sin^2(x)} dx \\ &u = \sin(x) \Rightarrow du = \cos(x) dx \\ &= 2 \int \frac{u du}{1 + u^2} \\ &v = 1 + u^2 \Rightarrow dv = 2u du \\ &= \int \frac{dv}{v} \\ &= \ln |1 + \sin^2(x)| + c \end{aligned}$$

References